

A Ground Penetrating Radar Survey of a Probable Cemetery Location at Fort Duffield, West Point, Hardin County, Kentucky

**By
Philip B. Mink, II
Jay Stottman**



**Letter Report
Kentucky Archaeological Survey
Jointly Administered By:
University of Kentucky
Kentucky Heritage Council
August 2014**

INTRODUCTION

In June of 2013 and July 2014, the Kentucky Archaeological Survey (KAS), on behalf of the Friends of Fort Duffield, conducted a ground penetrating radar (GPR) survey of a location that archival information indicated was the most probable locale of the Civil-War Era cemetery at Fort Duffield. This Civil War fortification is located near West Point, Kentucky (Figure 1). Following the processing of the data, soil-cores were used to evaluate several anomalies to determine if they represented grave shafts. The purpose of this survey was to attempt to locate evidence of any burials to assist in future interpretation and planning of this section of the Fort Duffield Park and Historic Site.

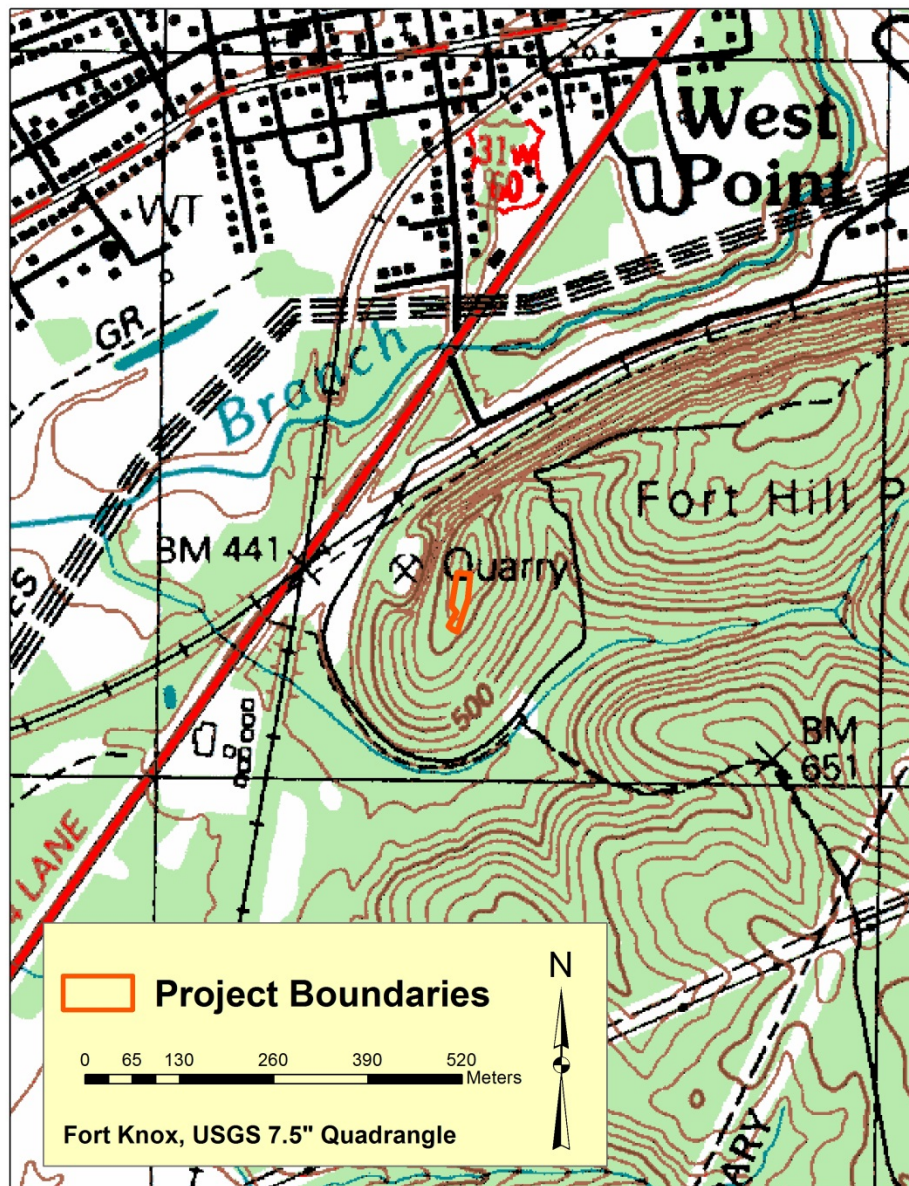


Figure 1 Location of Project Area on Fort Knox USGS 7.5" Quadrangle.

HISTORICAL BACKGROUND

The history of Fort Duffield has been extensively researched and reported by Richard A. Briggs (1999). This section will provide a brief summary of the fort's development and history to provide context for the geophysical investigations of the cemetery. It will primarily be focused on the history of the cemetery, as it is currently understood based on Briggs' work and information provided by members of the Friends of Fort Duffield.

The construction of Fort Duffield began in September of 1861 in response to a chance encounter and brief skirmish between Confederate cavalry and Union home guard troops at West Point. The realization that Confederate cavalry was encountered so close to Louisville created a small panic, as there were no formal troops there to protect it. Thus, construction of fortifications at West Point and across the River in Harrison County, Indiana was ordered to protect the city (Briggs 1999). The Home Guard troops hastily built crude earthworks in defense of the area. By late Fall several regiments had been ordered to the area in order to provide a more suitable defense of Louisville, including the 37th Indiana, the 1st and 18th Ohio, 9th Michigan, and the 16th and 28th Kentucky. The fort was planned by army engineer Colonel Nathaniel Michler, who also oversaw its construction. It was a three sided fort, with two sides protected by earthworks and the third rear side open to a 300 foot cliff. The fort was named after the father of the 9th Michigan's commanding officer, who as the ranking Colonel assumed command of the fort (Briggs 1999).

The cemetery at Fort Duffield was established when an epidemic that killed 61 members of the 9th Michigan broke out during the winter of 1861-62. They were buried side by side on the parade grounds of the fort located at the western tip of a hill according to Lieutenant Colonel Pankhurst in a letter to his family (Briggs 1999:41). By most accounts, the location of the cemetery was on the hill, specifically its western portion. As late as the 1970s, the location of the graves were denoted by a rusted sign and a series of field stones, indicating that at least in public memory the west end of the hill was the location of the burial ground (Briggs 1999). While the documentation on the location of the cemetery is fairly consistent the disposition of the soldier's remains is another matter. There are various reports that some of the soldier's remains were removed by family members and taken back to Michigan (Briggs 1999). According to National Cemetery records, 27 soldiers were removed from a trench at the top of Muldraugh Hill and reinterred at the National Cemetery in New Albany. The records indicate that they were removed from a trench 18 x 15 feet and 12 feet deep. Thus, it is unclear if and how many remains may be presently buried in the cemetery. However, it is likely that at least some remains have been removed and relocated to another cemetery.

GROUND PENETRATING RADAR SURVEY

INSTRUMENTATION

Geophysical methods of investigation were initially developed for the study of subsurface geological features. However, in recent years they have become more and more important for the detection of archaeological features (Smekalova et al 2005). Geophysical remote sensing technology has been applied to archaeological research in North America since the 1930s (Aitken 1961). During the 1980s, geophysical surveys in archaeology intensified as commercial equipment and software became more readily available, a trend that continues today. The increasing availability of commercial equipment amplified the quality and quantity of data collected and the more intensive usage by archaeologists resulted in theoretical and methodological advancements. One of the techniques that is beginning to be widely used in North America is ground penetrating radar (GPR). GPR is a near surface geophysical method that utilizes radar waves to measure the differential reflection properties of subsurface soil strata (Conyers and Goodman 1997). Radar waves (a type of electromagnetic radiation) are transmitted into the ground and reflect off buried discontinuities (e.g., rocks, architecture, graves, and pits). Measuring the rate of reflection in a study area allows a GPR user to search for anomalies within the area of interest (Conyers 2004).

When active, electromagnetic pulses are transmitted from a control unit. The pulses then propagate waves that penetrate the ground and reflect off buried discontinuities. The rate of time it takes for the wave to travel into the ground and reflect off the buried anomaly is measured and recorded by the GPR instrument (Conyers and Goodman 1997). A portion of the wave will continue further into the ground and continue to reflect off of discontinuities until finally the wave attenuates (the amplitude and/or intensity is too small to reflect back to the antenna). The electromagnetic properties of the soil described as relative dielectric permittivity (or the dialectic constant) can dramatically influence the depth and attenuation rate of the radar waves. High clay or salt content in soil, as well as water saturation (poorly-drained soil or subsurface water-pooling), will almost certainly result in poor GPR survey results. A more in-depth and technical discussion of GPR and archaeology can be found in Conyers and Goodman (1997) and Conyers (2004) and readers are encouraged to consult these sources for a more thorough discussion of the technical aspects of GPR.

The instrument used to conduct the survey was a Ramac GPR CU II Geo System with a 500 MHz shielded antenna (Figure 2). This instrument is capable of discriminating softball-sized objects (~0.04 meter) and has an optimal depth penetration of approximately 1–5 meters below the ground surface and a maximum penetration depth of approximately 3–10 meters below the surface. The CU II unit has a pulse repetition frequency of 100 kHz and the 500 MHz antenna emits broadband pulses for at least 500 MHz (250 MHz – 750 MHz).

Kentucky Archaeological Survey (KAS) archaeologists have previously used GPR on a variety of historic and prehistoric archaeological sites. One common use of GPR by KAS is to locate historic graves. For instance, at the State War Monument in the Frankfort Cemetery KAS utilized GPR capabilities to locate the graves of soldiers killed during the Mexican War and Civil War that had been covered over with a concrete and brick walkway in the early 1980's (Stottman and Pollack 2005).



Figure 2. Philip Mink and the Mala GPR unit on Grid 1 at Fort Duffield (Photo Courtesy of Connie Morris)

GPR METHODOLOGY

Two irregular shaped grids were established to collect data for this survey (Figure 3). Grid 1 was the smaller grid with a maximum width of 16 meters and maximum length of 24 meters and was located on the southwestern section of the “Original Burial Site” ridge. Grid 2 was the larger grid having a maximum width of 12 meters and a maximum length of 47 meters. It was contiguous with Grid 1 and extended northeast towards the current monument.

The GPR unit rests on a wheeled cart that is pushed by the operator in a back-and-forth method across the grid on 50 cm transects. The 50 cm transects were utilized to ensure sufficient and total subsurface coverage, as the radar waves spread in a conical fashion. The data collected during the field survey were processed at the University of Kentucky Archaeological Facility in GPR Slice v7.

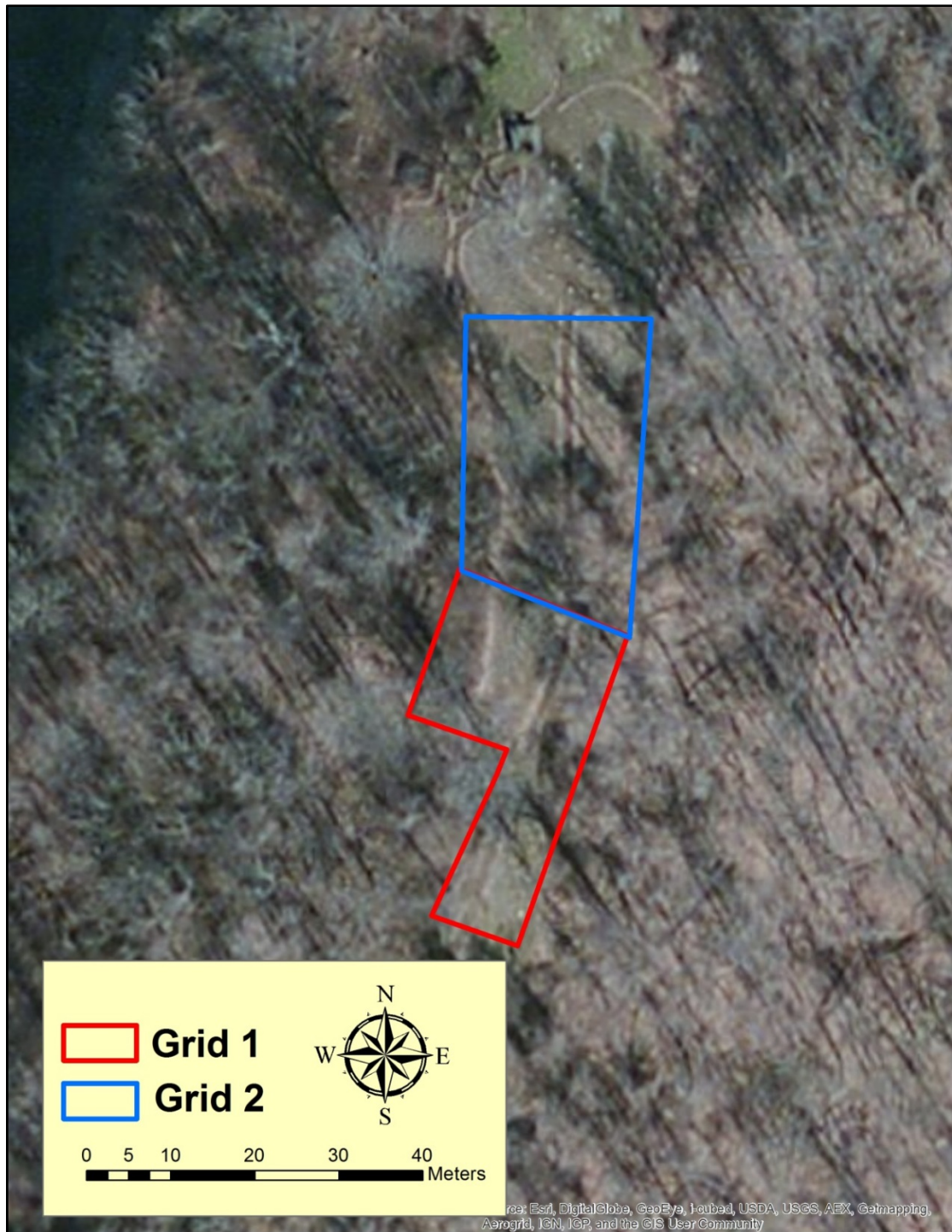


Figure 3. GPR Grids.

The GPR Slice results from the field survey are often presented as amplitude time slice maps. The amplitude slice map is similar to a medical CATSCAN, and can be thought of as a slice of the survey area at a particular depth. The data are initially sliced in a time unit called nanoseconds (1 billionth of a second) and later converted to a unit of space. The color scheme progresses from darker blue to light green, yellow, and red. Low reflections of soil density are represented by the darker (blue) of the color scheme, with higher reflection amplitudes increasing the brightness of the color scheme to green, yellow, then red. Thus, areas displayed as bright yellow, green, and red indicate subsurface anomalies, where the reflection of radar waves has been amplified, which suggests a cultural or natural discontinuity. The results are typically presented in a series of amplitude slice maps that can be thought of as plan maps of a site at a particular depth below the surface. Interpretations of these anomalies are based upon characteristics such as depth, size, shape and spatial distribution of the higher amplitude reflections. In some instances it is clear what the amplitude anomalies represent but in most cases some type of ground-truthing (soil-cores or shovel probes) will be required to determine whether the anomalies are archaeological or natural features.

GPR SURVEY RESULTS

It is important to reiterate that geophysical survey techniques simply measure variations in the physical properties of the earth and do not technically record the location of graves. Instead by examining the patterns present in the geophysical dataset the variations in the physical environment can be utilized to make inferences about unmarked burials (Kvamme 2008). A trained and experienced geophysical archaeologist employs their knowledge of the local archaeological record (including grave shafts), the existing geophysical anomaly database for the local region, and the depth, shape, size and intensity of recorded anomalies to interpret the patterns produced in the data and to identify possible unmarked graves.

Figure 4 illustrates the results of the GPR survey of Grid 1. The intense reflections (red, yellow, and green) running the length of the center of the grid represent compacted and disturbed soils associated with the existing pathway. This is most evident in slices a1-a6. The intense reflections (red, yellow, and green) evident in slices a1 and a2, with almost zero reflections on either side of the pathway (slices a4 and a5) are indicative of a modified landscape where leveling and filling have taken place. This may have been done intentionally or could have resulted when soil settled after burial trenches were excavated and re-excavated. The remaining slices (a7-a20) contain a variety of anomalies that required ground-truthing to determine if they could potentially represent grave shafts. Most of the anomalies are located along the eastern and western edges of the grid, and appear as high intensity isolated reflections that are the size, shape, and depth one would expect for a grave.

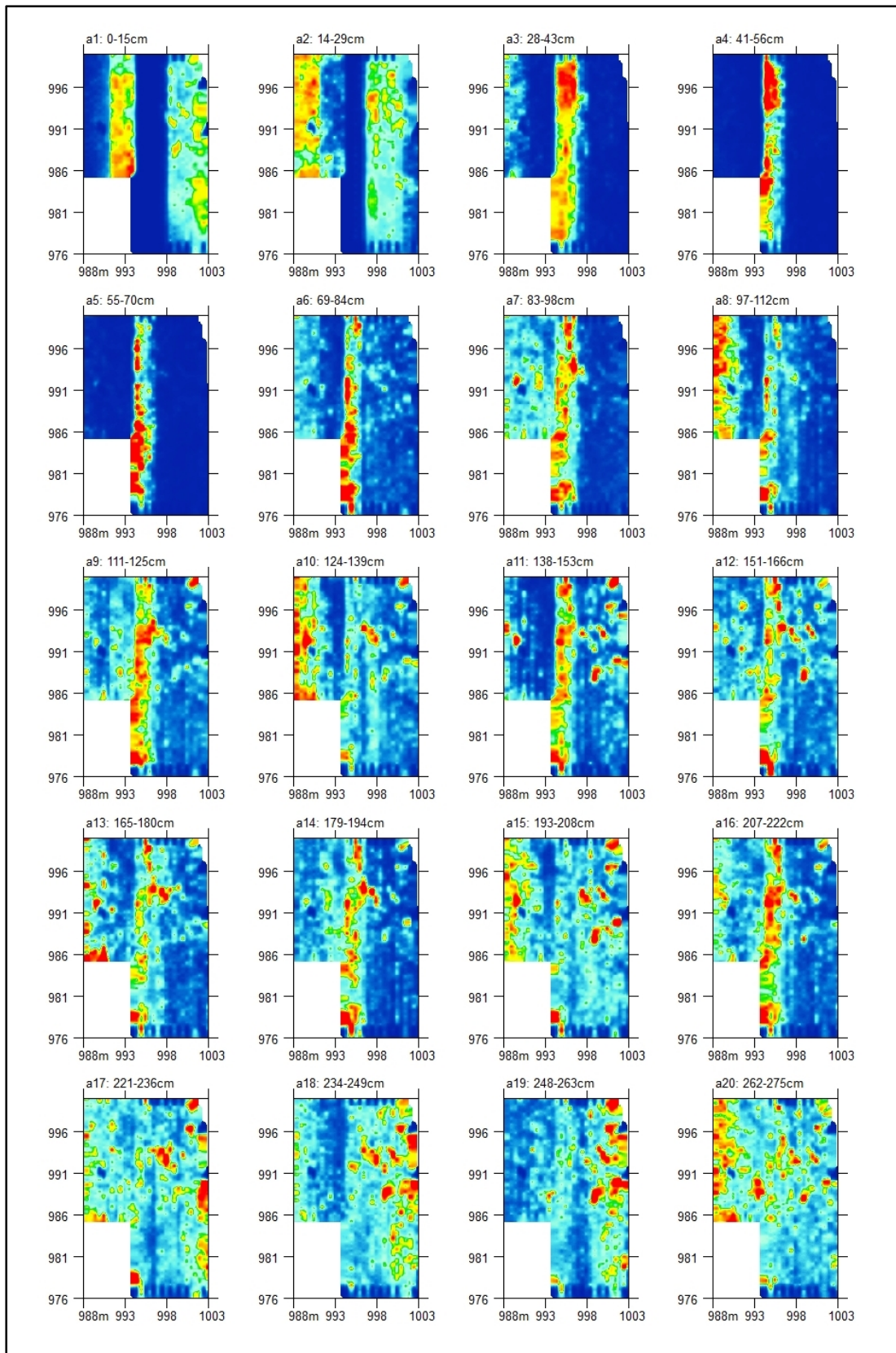


Figure 4. GPR results of Grid 1.

Figure 5 illustrates the results of the GPR survey of Grid 2, which is located closest to the monument. In this grid the existing pathway appears in the uppermost slices (e.g., a3, a6, and a7), as a medium intensity reflection (blue-green) that runs in an arc from the center of the grid toward the eastern edge and back to the center. The upper most slices (a1-a2) do show some disturbance in the upper 20 centimeters of the ground but the rest of the slices (a3-a20) seem to be fairly consistent. Beginning around 180-190 centimeters below ground (a14) and continuing down almost 3 meters an increase of high intensity reflective anomalies (red and yellow) appear along the eastern edge of the center of the grid. These are a bit deeper than one typically would expect for a grave at this time but archival information indicates that a trench up to 12 feet (3.66 meters) deep was dug for some of the burials.

SOIL CORES

In order to determine if the GPR anomalies were the result of natural processes (e.g., tree roots, soil development, or residual rocks) or cultural processes (e.g., grave shafts or trenches) ground-truthing with a three-quarter inch split spoon soil core was undertaken. Use of this type of core allows an investigator to examine and assess the soil stratigraphy in the vicinity of an anomaly without disturbing the surrounding ground surface.

The most promising anomalies detected with the GPR were targeted for evaluation to locate disturbed soils that could represent grave shaft/burial trench fill deposits. Such features would be identified by the presence of mottled unconsolidated soil. Core samples were taken to subsoil or to a maximum depth of 70 centimeters and were placed along seven transects (Figure 6).

Transects 1 and 2 were placed in the middle of Grid 1 to evaluate a series of linear anomalies (Figures 4 and 6). Transect 3 was placed along the west edge of Grid 1 where several additional anomalies were identified. Transects 4 and 5 were located along the east side of Grid 1 to investigate a scattering of anomalies around an area of stone (Figures 6 and 7). Cores associated with Transects 1-4 were placed at 2 meter intervals, while core intervals in Transect 5 ranged from 2 to 4 meters. Transect 6 was placed along the east side of Grid 2 and Transect 7 was located along the west side of Grid 2 (Figure 6). Soil cores were placed at 5 meter intervals along both transects.

The soil profile identified throughout most of the project area consisted of a 35 to 45 centimeter thick slightly mottled dark brown to brown silt loam topsoil that overlaid a yellow brown silt clay subsoil. Occasionally, the topsoil deposits extended to a depth of 60 centimeters. A mottled brown/yellow brown silt loam clay was identified in Transect 4 within the area of stone reported to be the location of the cemetery. In this area, the mottled soil extended to a depth of 50 centimeters before subsoil was encountered. In a few probes, subsoil was not reached due to the high density of stone (Figures 6 and 7). Mottled soil also was identified in two consecutive soil cores placed in the middle of Transect 6 (Figure 6). The mottled soil extended to a depth of 50 centimeters before subsoil was encountered.

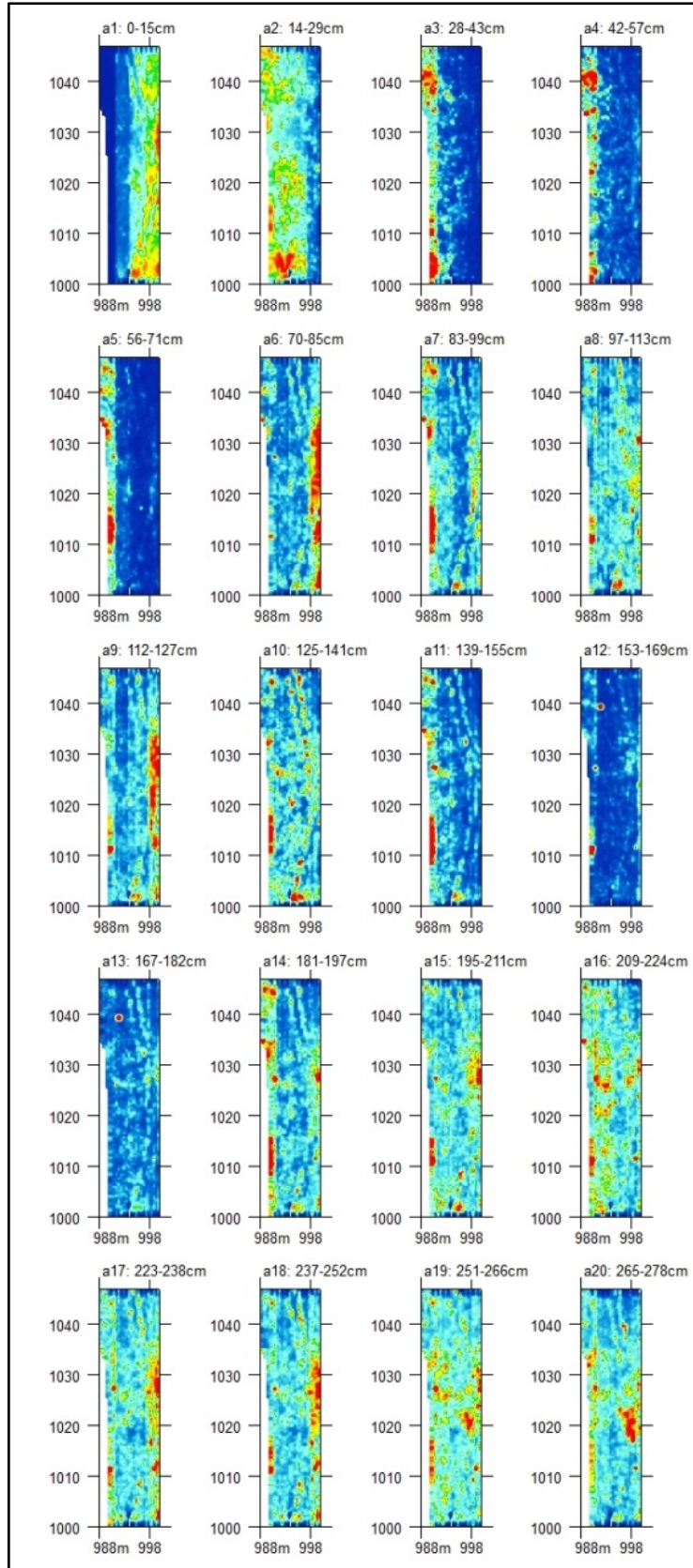


Figure 5. GPR results of Grid 2.

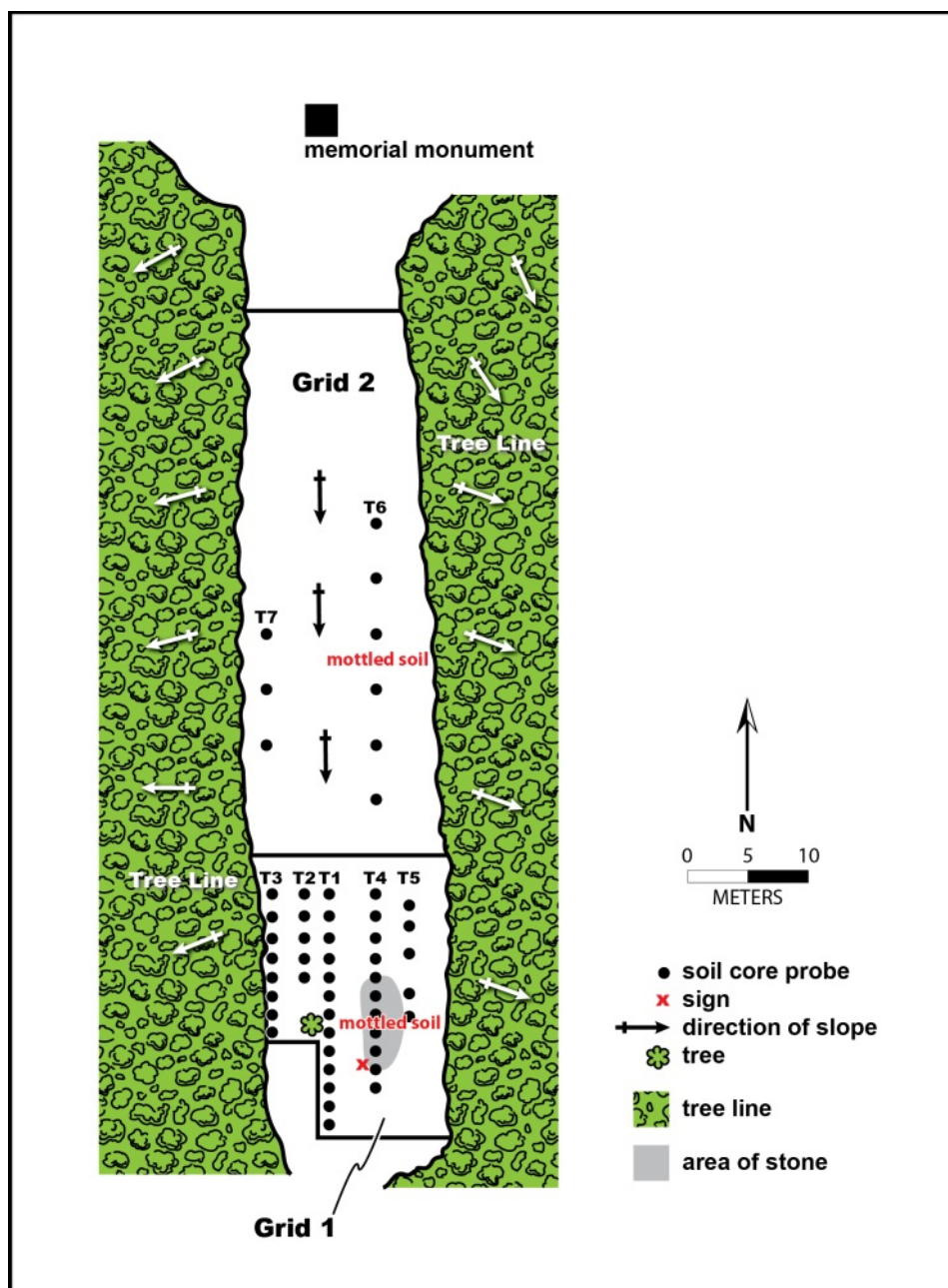


Figure 6. Soil Core Transect.

The results of the soil coring indicate that anomalies detected during the geophysical survey could not be definitively confirmed as graves or the location of the reported burial trench. The presence of mottled soil in the area of stone reported to be the location of the burials, however, does point to prior disturbance of this area. Since for the most part subsoil was encountered below the mottled soils, these deposits could not be definitively classified as grave shaft fill. Though the areas with mottled soils could not be identified as grave shafts with any degree of confidence, it is still possible that they represent the remnants of such features. The nature of the topography, erosion, and efforts to exhumate and rebury soldiers could potentially have altered the landscape and affected the GPR and soiling coring results.



Figure 7. The South End of Grid 1 Looking Southwest with Reported Burial Area to the Left.

SUMMARY AND RECOMMENDATIONS

In June of 2013 and July 2014, the Kentucky Archaeological Survey (KAS), on behalf of the Friends of Fort Duffield, conducted a ground penetrating radar (GPR) survey and soil-core evaluation of the most probable location of a Civil-War cemetery at Fort Duffield Park and Historic Site. The purpose of this study was to locate graves/burial trench associated with this cemetery, in order to provide information that could be used in future park interpretation and planning.

During the course of the geophysical survey, several anomalies were identified that based on their depth and shape potentially could represent graves. Evaluation of these anomalies determined that some represented disturbed soils that are consistent with grave shaft fill. That these areas were located near several field stones and just to the north of these stones was somewhat encouraging. Unfortunately, examination of soil core profiles could not confirm the presence of grave shafts or a burial trench with any degree of confidence. Despite these findings, there is substantial documentary evidence that indicates that the Fort Duffield cemetery is located within the project area, and thus that some of the anomalies detected during the geophysical survey are related to this cemetery cannot be totally discounted. Issues related to the site's topography, erosion, and attempts to remove and rebury soldiers may have altered the landscape in the area, effectively masking the location of individual graves and the burial trench. Given these findings it is likely that the Fort Duffield cemetery is located in the project area and where historically reported. It is thus recommended that park staff and volunteers continue to preserve and protect this area, and interpret it to the public as the possible location of a Civil War cemetery.

REFERENCES CITED

- Aitken, Martin J.
1961 *Physics and Archaeology*. Interscience Publishers, New York.
- Briggs, Richard A.
1999 *Fort Duffield West Point, Kentucky: The Saga of Fort Duffield Kentucky's Civil War Treasure*. Friends of Fort Duffield, West Point, Kentucky.
- Conyers, Larry B.
2004 *Ground-Penetrating Radar for Archaeology*. Alta Mira Press, New York.
- Conyers, Larry B. and Dean Goodman
1997 *Ground-Penetrating Radar: An Introduction for Archaeologists*. Alta Mira Press, New York.
- Handshoe, Donald and Christine Pappas
2009 *Geophysical Investigations at the Linden Grove Cemetery in Covington, Kentucky*. Letter report on file, Kentucky Archaeological Survey, Lexington.
- Kvamme, Kenneth L.
2008 Remote Sensing Approaches to Archaeological Reasoning: Pattern Recognition and Physical Principles, in *Archaeological Concepts for the Study of the Cultural Past*, edited by Alan P. Sullivan III, pp. 65-84. University of Utah Press, Salt Lake City.
- Murray-Wooley, Carolyn and Karl Raitz
1992 *Rock Fences of the Bluegrass*. University Press of Kentucky, Lexington.
- Stottman, M. Jay and David Pollack
2005 *Archaeological Investigations at the State Monument, Frankfort, Kentucky*. Report No. 104. Kentucky Archaeological Survey, Lexington.
- Somers, Lewis E.
1998 Geophysical Remote Sensing Survey of the Quartermaster Depot Dump at Fort Laramie National Historic Site. In *Archeology at the Fort Laramie Quartermaster Dump Area, 1994-1996*, edited by Danny. N. Walker, pp. 71-80. Division of Cultural Resource Selection Series No. 13. National Park Service, Rocky Mountain Region, Denver.
- Smekalova, Tatyana N., Olfert Voss, and Sergey L. Smekalov
2005 *Magnetic Survey in Archaeology*. Publishing House of Polytechnic University, St. Petersburg, Russia.